Student: Cajida | **Seed Number:** 3 | **Favorite Artist:** TUKI.

Execution Log (Complete, Unedited Program Output)

Test Sources: ['CAJIDA_T1', 'CAJIDA_T2', 'CAJIDA_T3', 'CAJIDA_T4', 'CAJIDA_T5']
Categories: ['TEMPERATURE', 'VOLTAGE', 'CURRENT', 'PRESSURE', 'SPEED']
Generated Values: [87, 91, 84, 85, 61, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90]

Total Records Generated: 28

Repeated Records: [{'source': 'CAJIDA_T1', 'category': 'TEMPERATURE', 'value': 87}, {'source': 'CAJIDA_T3', 'category': 'TEMPERATURE', 'value': 87}, {'source': 'CAJIDA_T4', 'category': 'VOLTAGE', 'value': 88}]
Distinct Source/Category Combinations: 25

Category Averages: {'TEMPERATURE': 88.71, 'VOLTAGE': 89.83, 'CURRENT': 88.8, 'PRESSURE': 89.0, 'SPEED': 84.2}
Source Averages: {'CAJIDA_T1': 82.5, 'CAJIDA_T2': 90.0, 'CAJIDA_T3': 89.5, 'CAJIDA_T4': 89.67, 'CAJIDA_T5': 90.0}

=== FINAL ORGANIZED AND SUMMARIZED TEST DATASET ===
Total Records: 28
Category Averages: {'TEMPERATURE': 88.71, 'VOLTAGE': 89.83, 'CURRENT': 88.8, 'PRESSURE': 89.0, 'SPEED': 84.2}
Source Averages: {'CAJIDA_T1': 82.5, 'CAJIDA_T2': 90.0, 'CAJIDA_T3': 89.5, 'CAJIDA_T4': 89.67, 'CAJIDA_T5': 90.0}

Assessment Data

#	Field	Entry
1	Generated Test Data	**Test Sources:** `['CAJIDA_T1', 'CAJIDA_T2', 'CAJIDA_T3', 'CAJIDA_T4', 'CAJIDA_T5']` **Categories:** `['TEMPERATURE', 'VOLTAGE', 'CURRENT', 'PRESSURE', 'SPEED']` **Raw Generated Values (25):** `[87, 91, 84, 85, 61, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90]`

2	Organized Test Data	**Total Records Generated:** 28 (25 base source-category pairs + 1 repeated pair + 2 additional category measurements)
3	Repeated/Distinct Data	[{'source': 'CAJIDA_T1', 'category': 'TEMPERATURE', 'value': 87}, {'source': 'CAJIDA_T3', 'category': 'TEMPERATURE', 'value': 87}, {'source': 'CAJIDA_T4', 'category': 'VOLTAGE', 'value': 88}] Distinct Combinations: 25
4	Numerical and Categorical Summaries	**Category Averages:** • `TEMPERATURE`: 88.71 • `VOLTAGE`: 89.83 • `CURRENT`: 88.8 • `PRESSURE`: 89.0 • `SPEED`: 84.2 **Source Averages:** • `CAJIDA_T1`: 82.5 • `CAJIDA_T2`: 90.0 • `CAJIDA_T3`: 89.5 • `CAJIDA_T4`: 89.67 • `CAJIDA_T5`: 90.0
5	Processed Results	Identification of 25 distinct combinations and the isolation of 3 duplicate entries from 28 total records.
6	Ordered Results	Dataset sorted lexicographically by `(source, category)`

7	Final Summary	`Total Records: 28`
		
		`Category Averages:`
		`{'TEMPERATURE': 88.71,`
		`'VOLTAGE': 89.83,`
		`'CURRENT': 88.8,`
		`'PRESSURE': 89.0, 'SPEED':`

		84.2} Source Averages: {'CAJIDA_T1': 82.5, 'CAJIDA_T2': 90.0, 'CAJIDA_T3': 89.5, 'CAJIDA_T4': 89.67, 'CAJIDA_T5': 90.0}

Final Output

=== FINAL ORGANIZED AND SUMMARIZED TEST DATASET ===
Total Records: 28
Category Averages: {'TEMPERATURE': 88.71, 'VOLTAGE': 89.83, 'CURRENT': 88.8, 'PRESSURE': 89.0, 'SPEED': 84.2}
Source Averages: {'CAJIDA_T1': 82.5, 'CAJIDA_T2': 90.0, 'CAJIDA_T3': 89.5, 'CAJIDA_T4': 89.67, 'CAJIDA_T5': 90.0}

Exercise 2: Student Performance Analyzer

Execution Log (Complete, Unedited Program Output)

Student Names: ['CAJIDA_S1', 'CAJIDA_S2', 'CAJIDA_S3', 'CAJIDA_S4', 'CAJIDA_S5']
Subjects: ['PROGRAMMING', 'MATHEMATICS', 'ELECTRONICS', 'PHYSICS', 'ENGLISH']
Generated Scores: [87, 91, 84, 85, 61, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90]

Total Records Generated: 27

Sample Classified Record: {'student': 'CAJIDA_S1', 'subject': 'PROGRAMMING', 'score': 87, 'category': 'GOOD'}

Repeated Records: [{'student': 'CAJIDA_S1', 'subject': 'PROGRAMMING', 'score': 87, 'category': 'GOOD'}]
Incomplete Records: [{'student': 'CAJIDA_S3', 'subject': 'PHYSICS', 'score': None, 'category': 'INCOMPLETE'}]

=== CONSOLIDATED PERFORMANCE REPORT ===
Total Records: 27
Student Averages: {'CAJIDA_S1': 82.5, 'CAJIDA_S2': 90.0, 'CAJIDA_S3': 90.0, 'CAJIDA_S4': 90.0, 'CAJIDA_S5': 90.0}
Subject Averages: {'PROGRAMMING': 89.0, 'MATHEMATICS': 90.2, 'ELECTRONICS': 88.8, 'PHYSICS': 89.0, 'ENGLISH': 84.2}

Assessment Data

#	Field	Entry
1	Generated Performance Records	27 total records (25 base + 1 repeated + 1 incomplete); scores: [87, 91, 84, 85, 61, 90 ×20]
2	Organized Student/Subject Data	Students CAJIDA_S1–S5 paired with subjects PROGRAMMING, MATHEMATICS, ELECTRONICS, PHYSICS, ENGLISH
3	Repeated/Incomplete Records	Repeated: CAJIDA_S1/PROGRAMMING (87); Incomplete: CAJIDA_S3/PHYSICS (None)
4	Performance Categories	Sample: CAJIDA_S1/PROGRAMMING = GOOD (87)
5	Student and Subject Summaries	Student Averages — **S1**: 82.5, **S2:** 90.0, **S3**: 90.0, **S4**: 90.0, **S5**: 90.0. Subject Averages — **PROGRAMMING**: 89.0, **MATHEMATICS**: 90.2, **ELECTRONICS**: 88.8, **PHYSICS**: 89.0, **ENGLISH**: 84.2
6	Ordered Results	Sorted alphabetically by student, then by subject

Final Output

=== CONSOLIDATED PERFORMANCE REPORT ===
Total Records: 27
Student Averages: {'CAJIDA_S1': 82.5, 'CAJIDA_S2': 90.0, 'CAJIDA_S3': 90.0, 'CAJIDA_S4': 90.0, 'CAJIDA_S5': 90.0}
Subject Averages: {'PROGRAMMING': 89.0, 'MATHEMATICS': 90.2, 'ELECTRONICS': 88.8, 'PHYSICS': 89.0, 'ENGLISH': 84.2}

Exercise 3: Transaction History Analyzer

Execution Log (Complete, Unedited Program Output)

Customers: ['CAJIDA_C1', 'CAJIDA_C2', 'CAJIDA_C3', 'CAJIDA_C4', 'CAJIDA_C5']
Products: ['NOTEBOOK', 'PEN', 'USB', 'HEADPHONES', 'CALCULATOR']
Quantities: [8, 9, 9, 7, 10, 2]

Generated Transaction Records: [{'customer': 'CAJIDA_C1', 'product': 'NOTEBOOK', 'quantity': 8}, {'customer': 'CAJIDA_C2', 'product': 'PEN', 'quantity': 9}, {'customer': 'CAJIDA_C3', 'product': 'USB', 'quantity': 9}, {'customer': 'CAJIDA_C4', 'product': 'HEADPHONES', 'quantity': 7}, {'customer': 'CAJIDA_C5', 'product': 'CALCULATOR', 'quantity': 10}, {'customer': 'CAJIDA_C1', 'product': 'NOTEBOOK', 'quantity': 8}]

Total Transaction Value: 9985
Repeated Transactions: [{'customer': 'CAJIDA_C1', 'product': 'NOTEBOOK', 'quantity': 8, 'value': 400}]
Distinct Transactions: [{'customer': 'CAJIDA_C1', 'product': 'NOTEBOOK', 'quantity': 8, 'value': 400}, {'customer': 'CAJIDA_C2', 'product': 'PEN', 'quantity': 9, 'value': 135}, {'customer': 'CAJIDA_C3', 'product': 'USB', 'quantity': 9, 'value': 2250}, {'customer': 'CAJIDA_C4', 'product': 'HEADPHONES', 'quantity': 7, 'value': 5600}, {'customer': 'CAJIDA_C5', 'product': 'CALCULATOR', 'quantity': 10, 'value': 1200}]

Customer Summaries: {'CAJIDA_C1': 800, 'CAJIDA_C2': 135, 'CAJIDA_C3': 2250, 'CAJIDA_C4': 5600, 'CAJIDA_C5': 1200}
Product Summaries: {'NOTEBOOK': 800, 'PEN': 135, 'USB': 2250, 'HEADPHONES': 5600, 'CALCULATOR': 1200}

=== FINAL CONSOLIDATED TRANSACTION REPORT ===
Total Transactions: 6 | Total Value: 9985

Assessment Data

#	Field	Entry
1	Generated Transaction Records	6 records: **C1/NOTEBOOK/8**, **C2/PEN/9**, **C3/USB/9**, **C4/HEADPHONES/7**, **C5/CALCULATOR/10**, **C1/NOTEBOOK/8** *(repeated)*
2	Organized Transaction Data	Each record links customer, product, quantity, and computed value
3	Repeated/Distinct Transactions	Repeated: **CAJIDA_C1/NOTEBOOK** (value 400); Distinct: the other 5 unique pairs
4	Transaction Values	**C1/NOTEBOOK**: 400, **C2/PEN**: 135, **C3/USB**: 2250, **C4/HEADPHONES**: 5600, **C5/CALCULATOR**: 1200
5	Customer Summaries	**C1**: 800, **C2**: 135, **C3**: 2250, **C4**: 5600, **C5**: 1200
6	Product Summaries	**NOTEBOOK**: 800, **PEN**: 135, **USB**: 2250, **HEADPHONES**: 5600, **CALCULATOR**: 1200
7	Ordered Results	Sorted alphabetically by customer, then product
8	Overall Transaction Summary	Total Transactions: 6, Total Value: 9985

Final Output

Final Consolidated Transaction Report

Customer Summaries: {'CAJIDA_C1': 800, 'CAJIDA_C2': 135, 'CAJIDA_C3': 2250, 'CAJIDA_C4': 5600, 'CAJIDA_C5': 1200}

Product Summaries: {'NOTEBOOK': 800, 'PEN': 135, 'USB': 2250, 'HEADPHONES': 5600, 'CALCULATOR': 1200}

=== FINAL CONSOLIDATED TRANSACTION REPORT ===
Total Transactions: 6 | Total Value: 9985